

César Alfonso Fierro Vizcarra

Mechatronic Engineer Robot programmer

📍 Chihuahua, Mexico

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ROBOTS

- FANUC
- KUKA
- ABB

PROGRAMMING STANDARDS

- GM (GRS Certified)
- VASS (VASS 5 Certified)
- FORD
- TESLA
- BMW

TECHS & APPLICATIONS

- Welding (SPOT & MIG)
- Sealer & Glue
- Material Handling
- IRS Vision (FANUC)

SOFTWARES

- ROBOGUIDE (FANUC)
- ROBOTSTUDIO (ABB)
- KUKA WORK VISUAL
- RS Logix 5000
- Factory Talk View
- PRONETA

PROJECTS PORTFOLIO

TESLA Empire Warehouse – 2026

Fanuc robot commissioning, PROFINET Communication setup, IO Link, Roboguide simulation for new lines, commissioning and configuration of Balluff and Festo sensors (pressure and distance), 3DV Vision setup and programming.

FORD (Irapuato Electric Powertrain Center) – 2025-2026

ABB Robot commissioning, robot programming, material handling application. FORD IFCW framework. Fanuc robot commissioning, programming routines, 3DV Vision setup and programming.

TESLA GIGAFACTORY Austin - 2025

FANUC Robot programming, DCS, integration, commissioning new lines. Path planning and programming. Cycle time improvements.

BMW Spartanburg - 2025

Kuka robot programming, path and search programming, safety commissioning. Correct robot projects. PRONETA, SAS, WORK VISUAL.

Battery Assembly Lines Integration - GM Plant – Spring Hill, TN, USA. 2023-2024

Fanuc Robot programming, first commissioning, Master robots, MH first setups, Sealer applications first set up, path programming, cycle time and lines improvements, production support. With ATS, Atom Tech, and Talos Automation.

KUKA METRO KMP – GM SAN LUIS POTOSI 2022-2023

Fanuc Robot programming, first commissioning, path programming. Spot Welding, Material Handling.

KUKA LIVONIA – GM CAMI. 2021

Fanuc Robot Programming, first commissioning, load offline, path programming, cycle time. Spot Welding, MIG Welding, Material Handling, Sealer.

Ford Stamping and assembling Plant. 2020-2021

KUKA Safety commissioning, robot programming, path and logic programming, shift support in presses line, with BotsIA Systems and JIER NA.

Ford Chihuahua Engine Plant. 2020-2021

ABB, robot programming, shift support, cycle time and line improvements. With BotsIA Systems, Essam Controls and MG Marposs.

AUDI Brussels – E-tron – Electric Vehicle. 2019-2020

Fanuc Robot programming in Battery montage lines. Standard VASS 6, shift support, cycle time and line improvements. With RAE Robotics and VH Robotics.

Volkswagen Chattanooga. 2018-2019

Fanuc Robot programming, standard VASS 5, integration, and commissioning new lines. With RAE Robotics and Dressler Automation.

Volkswagen Chattanooga. 2018-2019

Fanuc Robot programming, shift support, cycle time and line improvements. With RAE Robotics.

Tesla Plant Fremont California – Model 3. 2017-2018

KUKA KRC4 Robot programming, Safe Operation, integration, commissioning new lines for Model 3. With RAE Robotics and VDL.

Volkswagen Changchun China. 2017

KUKA KRC4 Robot programming; load, and test offline programs. With RAE Robotics and EngroTec.

ACADEMIC PROJECTS

IA Predictive Maintenance System – Thesis Project

Developed a machine learning system in Python to predict failures in industrial equipment using historical operational data. Implemented LSTM neural networks with TensorFlow and Keras. Used Pandas and NumPy for data wrangling, and Matplotlib for visualization. Aimed at reducing downtime and enabling preventative maintenance strategies.

Programming and optimization for industrial robotics paths with genetic algorithms

Developed a system of path programming, using genetic algorithms with Python to find optimal movements on industrial robots. Focused on reducing time programming times on projects.